2 Week 2: Cellular Evolution & 5G/6G Architectures

2.1 Tracking the architectural leap from legacy cellular generations to cloud-native frameworks.

2.1.1 The Generational Evolution (1G to 4G LTE)

The historical trajectory of cellular communications is characterized by a shift from rigid, hardware-centric analog systems to flexible, software-defined, and fully internet protocol (All-IP) flat architectures. Each generation introduced fundamental modifications to access methodologies, core topologies, and spectral efficiency.

A. **First to Third Generation (1G - 3G Foundations):**

(a) **1G (Analog Voice Era):** Deployed using Analog Advanced Mobile Phone System (AMPS) standards utilizing Frequency Division Multiple Access (FDMA). Systems were strictly circuit-switched, bound by low capacity, highly vulnerable to cross-talk, and entirely lacked cryptographic security mechanisms.

(b) **2G (Digital Voice Transition):** Shifted the physical layer to digital transmission using Time Division Multiple Access (TDMA/GSM) and Code Division Multiple Access (CDMA/IS-95). Introduced the Base Station Subsystem (BSS) and the Network Switching Subsystem (NSS), enabling the first commercial implementation of cryptographic authentication and text-based short message services (SMS).

(c) **3G (Broadband & IP Integration):** Engineered around Universal Mobile Telecommunications System (UMTS) using Wideband CDMA (WCDMA). This generation introduced the split between the circuit-switched core (for legacy voice) and the packet-switched core (for data routing via GPRS Support Nodes), facilitating the transition toward mobile internet access.

B. **Fourth Generation (4G LTE & All-IP Flat Architecture):** 4G Long-Term Evolution (LTE) fundamentally decoupled the cellular infrastructure from circuit switching by implementing an end-to-end All-IP Network (AIPN) architecture.

a **Physical Layer Shift:** Replaced CDMA with Orthogonal Frequency Division Multiple Access (OFDMA) for downstream paths to eliminate intra-cell interference and improve spectral density.

b **Evolved Packet Core (EPC):** Flattened the core network topology by eliminating the legacy Radio Network Controller (RNC). Macro base stations, designated as eNodeB (Evolved NodeB), assumed direct control over radio resource management and interface directly with the core via an IP routing backhaul.

Table 1: GENERATIONAL MILESTONES

Generation	**Primary Access Technology**	**Core Network Paradigm**	**Peak Theoretical Data Rate**	**Key Architectural Constraint**
1G	FDMA	Circuit-Switched (Analog)	2.4 kbps	Severe spectral inefficiencies; zero cryptographic isolation.
2G	TDMA / CDMA	Circuit + Basic Packet (Digital)	64 kbps (GPRS/EDGE)	Rigidly bound to voice-centric, narrow-band slots.
3G	WCDMA	Split Core (CS for Voice / PS for Data)	2 Mbps – 42 Mbps	Dual-core processing increases operational complexity and latency.
4G LTE	OFDMA / SC-FDMA	Evolved Packet Core (Pure All-IP)	100 Mbps – 1 Gbps	Hardware-dependent core elements; cannot dynamically adapt to heterogeneous IoT demands.

2.1.2 5G Architectural Paradigms: Non-Standalone (NSA) vs. Standalone (SA)

The commercial deployment of 5G introduces a fundamental architectural bifurcation. To mitigate immediate capital expenditure, operators deploy Non-Standalone (NSA) overlays before transitioning to pure, cloud-native Standalone (SA) topologies.

Table 2: DIRECT STRUCTURAL COMPARISONS (NSA VS. SA)

Technical Property	**Non-Standalone (NSA) Framework**	**Standalone (SA) Framework**
Core Network Dependency	Relies entirely on the legacy 4G Evolved Packet Core (EPC).	Operates on the native, cloud-born 5G Core (5GC).
Radio Access Configuration	Uses E-UTRA-NR Dual Connectivity (EN-DC).	Pure Next-Generation Radio Access Network (NG-RAN) via gNodeB.
Control Plane Anchor	Signaled and anchored via the existing 4G eNodeB infrastructure.	Complete control plane routing managed natively through the 5G gNodeB.
End-to-End Latency Profile	Restricted by legacy 4G EPC transport processing ($\sim$15–25 ms).	Ultra-low deterministic latency enabled by automated 5GC paths ($\leq$ 1 ms).
Advanced Feature Support	Incapable of executing dynamic native network slicing or edge integration.	Fully enables dynamic Network Slicing and Edge Computing (MEC).



(a) Non-Standalone Framework (b) Standalone Framework

Figure 1: Radio Access Network Structural Topology Comparison

2.1.3 The 5G Service-Based Architecture (SBA) and Core Network Functions

The 5G Core (5GC) completely breaks away from point-to-point hardware interfaces. It is structured as a Service-Based Architecture (SBA) where control plane entities are decoupled into stateless, modular, and containerized software components known as Network Functions (NFs). These NFs communicate via standard HTTP/2 (RESTful APIs) over a centralized Service-Based Interface (SBI) bus.

Core Architectural Separations (CUPS Principle)

5G strictly enforces the Control and User Plane Separation (CUPS) protocol. This allows user data routing paths to be scaled completely independently from session control mechanics.

1. **User Plane Function (UPF)**: The UPF is the solitary control-plane-free component executing packet data processing within the user plane. It functions as a highly optimized, high-throughput gateway responsible for packet routing, forwarding, inspection, quality of service (QoS) enforcement, and serves as the anchor point for external Packet Data Networks (PDNs). Because it is decoupled from the control plane, a UPF can be localized directly at the network edge to minimize propagation loop latencies.

2. **Access and Mobility Management Function (AMF):** The AMF acts as the primary termination point for the control plane signals from the user equipment (UE) and the radio access network. It manages terminal registration status, connection setup, security context distribution, and mobility management (handovers and tracking area updates). It does not manage session routing details directly; instead, it offloads session tracking to the SMF.

3. **Auxiliary Network Functions (NFs):**

    a. **Session Management Function (SMF):** Responsible for IP address allocation, session establishment, modification, and direct management of the UPF routing matrices.

    b. **Unified Data Management (UDM):** A cloud repository storing subscriber credentials, access profiles, and roaming permissions (the 5G equivalent of the legacy HSS).

    c. **Network Slice Selection Function (NSSF):** Dynamically maps user equipment requests to the specific logical target slices authorized for their subscription profiles.

2.1.4 Network Slicing Mechanics

Network Slicing is a structural paradigm that permits a physical network infrastructure to be logically partitioned into multiple, isolated, end-to-end virtual networks. Each slice is custom-tailored with unique performance characteristics (bandwidth, latency, reliability) without creating physical hardware dependencies.

The Enabling Pillars

1. **Software-Defined Networking (SDN):** Decouples the forwarding plane from the routing control logic, enabling programmatic control over network traffic paths.
2. **Network Functions Virtualization (NFV):** Converts specialized core hardware into modular software containers running on commercial off-the-shelf (COTS) cloud servers.
   The Three Standardized Slice Templates
   a. **Enhanced Mobile Broadband (eMBB):** Optimized for massive data throughput and spectral efficiency (e.g., streaming high-definition media across the Kirkuk University campus center).
   b. **Ultra-Reliable Low-Latency Communications (URLLC):** Engineered for sub-millisecond latency bounds and deterministic packet delivery probabilities ($\geq 99.999\%$). This slice utilizes optimized, shorter frame structures in the physical layer to support time-critical applications like autonomous vehicular arrays or automated smart grid grids.
   c. **Massive Machine-Type Communications (mMTC):** Optimized to maintain simultaneous, low-power connections for tens of thousands of low-data-rate sensor devices per square kilometer without over-allocating signaling channels.

2.1.5 Vision and Early Paradigms for 6G Architectures

As 5G reaches architectural maturity, early structural definitions for Sixth-Generation (6G) frameworks introduce advanced paradigms designed to unify communication, computing, and environmental sensing.

1. **Spectrum Expansion into the Terahertz (THz) Band:** 6G scales beyond millimeter waves into Sub-THz and THz spectral blocks (0.1 THz to 10 THz). While offering unmatched transmission channels, THz bands suffer from profound atmospheric molecular absorption and extreme free-space path loss. To bypass these physical limitations, 6G relies heavily on Intelligent Reflecting Surfaces (IRS)—metamaterial arrays embedded on structures like campus facility walls that programmatically re-radiate and steer EM wavefronts around blind non-line-of-sight spots.
2. **AI-Native Air Interfaces:** Replaces static, hard-coded physical layer blocks (such as channel coding, equalization, and modulation mappings) with adaptive, deep-learning neural models. The air interface continuously self-optimizes its operational variables based on real-time micro-fluctuations observed across the local radio environment.
3. **Space-Air-Ground-Sea Integration (Non-Terrestrial Networks - NTN):** While legacy networks remain tethered to terrestrial cell towers, 6G incorporates a multi-tier routing architecture that seamlessly merges low-earth orbit (LEO) satellite constellations, high-altitude platform systems (HAPS), and terrestrial nodes into a unified routing grid, providing global, high-density coverage across all spatial planes.

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